Full-Stack Development Challenge

SafeHaul — Smart Truck Route Planning

Build a React + Django web app that helps a truck driver choose the safest route based on weather, load weight, and travel time.

Requirements

The user enters:

- Origin & destination
- Departure date/time
- Load weight

The app should:

- Generate 3 route options using a routing API.
- Sample weather every 10 / 25 / 50 miles, using the truck's estimated arrival time at each checkpoint.
- Calculate weather risk using the provided risk/load rules.
- Recommend the best route, balancing weather risk and travel time.
- Show all routes, checkpoints, risk levels, ETAs, and route summaries on a map.
- Show a weather heatmap along the trip corridor with a 0–48 hour forecast slider.

You may use third-party APIs for routing, geocoding, and weather. You do not need to build a routing engine.

Risk Rules

Use these thresholds to classify weather at each checkpoint:

Condition	Low	Moderate	High	Severe	No Travel
Wind	<25 mph	25–34	35–44	45–54	≥55 mph
Rain	<0.10 in/hr	0.10–0.25	0.25–0.50	0.50–1.00	>1.00
Snow	<0.5 in/hr	0.5–1.0	1.0–2.0	2.0–3.0	>3.0

Load rules:

- ≥55 mph → No Travel for any load
- 45–54 mph + >30,000 lb → No Travel
- 35–44 mph + >40,000 lb → Severe

Route Recommendation

Prefer:

- 1. Fewest Severe miles
- 2. Fewest High miles
- 3. Lowest average risk
- 4. Shortest travel times

Deliverables

- GitHub repository
- Hosted application
- 2–3 minutes Demo Video

Deadline: 05 November 2026

Prioritize correctness and clarity over extra features.